Sourabh Bose

Email: sourabh.bose@wichita.edu Phone: (+1) 817-881-2284
LinkedIn: linkedin.com/in/sourabhbose GitHub: github.com/smokingelephants

Skills

Research Interests • Machine Learning, Data Analytics, Artificial Intelligence, Generative AI, Hierarchical Reinforcement Learning, Natural Language Processing

Programming Over 10000 lines: • Python, C, C++, Matlab, Java
Tools and libraries • Tensorflow, PyTorch, Caffe, Theano • OpenAI framework • AWS EC2, Azure, Google Cloud • Robot Operating System (ROS)

Education

2013 – 2019	University of Texas at Arlington Ph.D., Computer Science	Arlington, TX
2010 – 2012	University of Calcutta MS, Computer Science	Calcutta, WB

Work Experience

2019 – present	Wichita State University Assistant Teaching Professor Graduate Advisor, School of Computing	Wichita, KS
2013 – 2019	University of Texas at Arlington Graduate Research Assistant	Arlington, TX

Research Experience

- **MDP Auto-encoder** — Learn behaviorally grounded, concise representations for decision making problems with high dimensional observed space.
- **Imitation Learning** — Learning to imitate human behavior by enabling different robots to perform new tasks by mapping human demonstrations to robot space.
- **Adaptive Neural Network Classifier** — Incrementally build neural network classifiers that adapt to changing task requirements, applicable for big data or streaming data sources.
- **Learning Structured Representations** — Enforce structural constraints on neural network features, using policy gradient algorithms.
- **Semi-unsupervised clustering** — Learning dynamic representations for context-aware clustering problems.
- **Other Work**
  Dissertation — Learning Representations Using Reinforcement Learning
  Ph.D. Committee: Manfred Huber, Gergely Zaruba, Farhad Kamangar, Vassilis Athitsos
  Supported by NSF Grant: 1551312
  Masters Project — Multi-Agent Intelligent system aided Path finder – using Ant Colony Optimization and Bee Colony optimization, locate, track and calculate near-optimal path to a static/dynamic object.

Publications

- S. Bose and M. Huber, "MDP Autoencoder," in IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2019.
- S. Bose and M. Huber, "Training neural networks with policy gradient," in International Joint Conference on Neural Networks (IJCNN). IEEE, 2017.
- S. Bose and M. Huber, "Incremental learning of neural network classifiers using reinforcement learning," in IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2016.
- S. Bose and M. Huber, "Semi-Unsupervised Clustering Using Reinforcement Learning," in The Twenty-Ninth International FLAIRS Conference, 2016.
- S. Bose, "Learning Representations Using Reinforcement Learning," The University of Texas at Arlington; 2019.

Teaching Experience

2019 to present — Wichita State University — Wichita, KS
Assistant Teaching Professor

- Advanced Algorithms and Analysis (CS721)
- Algorithms and Applications on Graphs (CS797P) : *Integrated competitive programming like LeetCode and CodeChef-style coding problems to enhance students' problem-solving skills. Introduced advanced concepts like spectral graph theory and tropical semiring methods, making these abstract topics accessible through hands-on activities and real-world applications tailored for both graduate and undergraduate students.*
- Reinforcement Learning (CS898BG) : *Updated the course to reflect cutting-edge advancements in the field, including the use of large language models (LLMs) and Reinforcement Learning with Human Feedback (RLHF) training techniques.*
- Machine Learning (CS770/CS697AB)
- Introduction to Programming (CS211/CS211L)
- Introduction to Database Systems (CS665)
- Assembly Language Programming (CS238)
- Programming Paradigms (CS410/CS510)
- Honors Independent Research or Creative Activity (HNRS 485)

2019 to present — Wichita State University — Wichita, KS
Graduate Advisor

- Advised up to 1,400 graduate students across MS Computer Science, MS Computing, MS Data Science, and MS Computer Networking programs, assisting both international and domestic students with academic guidance, program selection, admissions, and onboarding.
- Contributed to curriculum design as part of academic committees, aligning course development with industry trends and academic standards.

WSU Recognitions

2023 — Nominated for Teaching Excellence Award, WSU

Professional Activities

Program Committee Memberships

– PerHealth 2020: 5th IEEE PerCom Workshop on Pervasive Health Technologies

Conference Reviewing:

– IEEE Transactions on System, Man, and Cybernetics
– PerHealth 2020

Professional Affiliations:

– IEEE Member
– AAAI Member